

Reconfigurable Power Divider and Combiner with Variable Power Ratio

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ABSTRACT — This paper presents the design of a power divider and combiner with variable power ratios. The essential idea is to use the operating principle of a coupled-line coupler. The couple-line coupler, which has control switches at the coupled and the isolation ports, offer several different operating conditions depending on the states of the switches. By means of changing the terminating-state of the coupled and the isolation ports, SHORT or OPEN, the power ratio becomes 1:1 or 1:0. The input power is either equally divided into two output ports or delivered to the one output port exclusively.

Index Terms — Wilkinson power divider, power ratio, characteristic impedance, coupled-line, hybrid.

I. INTRODUCTION

With rapidly increasing demand of the recent mobile communication systems, the stability of the service becomes very important issue. To improve quality of the service, an establishment of service stations is also increasing dramatically. Furthermore, it is more important to keep the service working continuously when the systems have some serious problem. For this reason, it is necessary to control power flow.

In radio frequency (RF) and microwave systems, power divider/couplers essentially divide or combine target signals at a given power dividing/combining ratio as an n-port network. For instance, in a 3 dB power divider case, all input power is equally divided into two output ports. [1] But, it is difficult to design a power divider/combiner with variable ratio, from 1:1 to 1:0 or from 1:0 to 1:1 using existing conventional power divider/combiner schemes

In this paper, a reconfigurable power divider/combiner, which has two different power ratios, is proposed. Transmission lines in the proposed scheme are implemented using a coupled line or a hybrid instead of utilizing conventional microstrip lines. Depending on the status of the coupler and hybrid, which have control switches at the coupled and isolation ports, the proposed divider/combiner change the power distribution rate from 1:1 to 1:0.

II. DESIGN THEORY

Fig. 1 shows the structure of a conventional m:n Wilkins power divider.

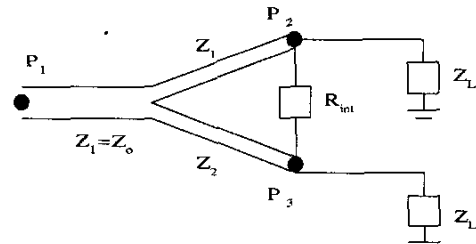


Fig. 1 m:n Wilkinson power divider

In this m:n Wilkinson power divider, the input power shown in Fig.1 is split to two output ports at the m:n power dividing ratio with the same amount of phase. Wilkinson power dividers with an arbitrary power dividing ratio is expressed as follows in equations (1) through (5). [2-3]

$$Z_1 = Z_o \sqrt{\frac{1+K^2}{K^3}} \quad (1)$$

$$Z_2 = Z_o \sqrt{K(1+K^2)} \quad (2)$$

$$R_{int} = Z_o \left(K + \frac{1}{K} \right) \quad (3)$$

$$Z_{L1} = Z_o K \quad (4)$$

$$Z_{L2} = Z_o / K \quad (5)$$

In this work, the proposed reconfigurable power divider has two different power dividing ratio, 1:1 and 1:0. In one operating mode, all input power is equally divided to two output ports like the conventional Wilkinson power divider. In the other state, all input power propagates to one output port exclusively, which is not realizable by using the conventional Wilkinson divider. In Fig. 2, the proposed new power dividing/combining scheme is shown.

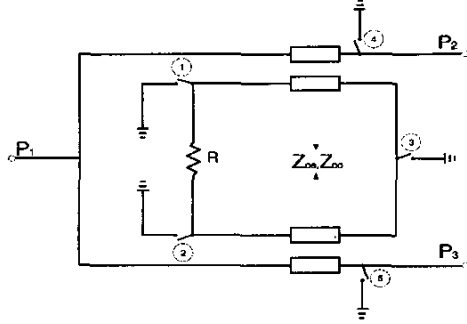


Fig. 2 Reconfigurable power divider
(Couple-line structure)

When switches, 1 ~ 5 are turned off, the proposed power divider behaves as a conventional 3dB Wilkinson power divider. However, when switches, ① 1 ~ 3, and 5 are turned on, all input power directs to P₂.

First the even-mode analysis of the scheme shown in Fig. 2 reveals that coupled-line impedance is given by equation. (6). [2-3]

$$\frac{Z_{oe} + Z_{oo}}{2} = Z_{oc} \quad (6)$$

In Fig. 2 when the switches, ① 1 ~ 3 and 5 are connected to the ground, the coupled-line impedance becomes as follows, equation (7):

$$\frac{2Z_{oe} \times Z_{oo}}{Z_{oe} + Z_{oo}} = Z_{sc} \quad (7)$$

From these equations (6) and (7), the even- and odd-mode characteristic impedances of the coupled-line shown in Fig 2 can be calculated.

$$Z_{oo} = \sqrt{\frac{Z_{sc} Z_{oc}}{x}} \quad (8)$$

$$Z_{oe} = x Z_{oo} \quad (9)$$

$$\text{where, } x = \left\{ 2 \left(\frac{Z_{oc}}{Z_{sc}} \right) - 1 \right\} + \sqrt{\left\{ 2 \left(\frac{Z_{oc}}{Z_{sc}} \right) - 1 \right\}^2 - 1}$$

Isolation resistance can be evaluated from the odd-mode analysis of the scheme in Fig. 2.

$$R = \frac{2 \left(\frac{2Z_{oe} \cdot Z_{oo}}{Z_{oe} - Z_{oo}} \right)^2}{Z_o} \quad (10)$$

As shown in Table 1, due to a large difference of the odd-mode and even-mode impedances, it is difficult to implement a quarter-wave length transmission line of power divider using the coupled-line. Therefore, the coupled-line was transformed to hybrids as shown in Fig. 3.

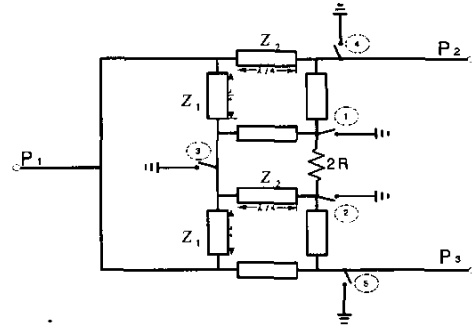


Fig. 3 Reconfigurable power divider with hybrids
(Hybrid structure)

The transformation from the coupled-line to the hybrid was done as follows, equations (11) through (16). [4]

$$10^{S_{21}/10} = A(\text{Direct Power}) \quad (11)$$

$$10^{S_{41}/10} = B(\text{Coupled Power}) \quad (12)$$

$$K^2 = \frac{P_{CP}}{P_{DP}} = \frac{B}{A} \quad (13)$$

$$Z_o = \sqrt{Z_{oe} \cdot Z_{oo}} \quad (14)$$

$$Z_1 = \frac{Z_o}{K} \quad (15)$$

$$Z_2 = \frac{Z_o}{\sqrt{(1 + K^2)}} \quad (16)$$

Where, K^2 =power ratio of branch-line hybrid

In equations (11) and (12), the S_{21} and S_{41} are S-parameters of the coupled-line at the direct and the coupled ports, respectively. Note that Z_o describes the characteristic impedance of the transmission line in the coupled-line and hybrids.

III. SIMULATION AND MEASUREMENT

Using the proposed method, the characteristic impedance of each transmission line is designed. For this simulation, Ansoft-Designer was used. Reconfigurable power divider with variable power ratio 1:1 and 1:0 was simulated, designed, and tested. For the circuit implementation, the 31 mil-thick RT/Duroid with dielectric constant of 2.2 was used.

Table.1 shows characteristic impedances of a coupled-line and a power divider at the given power-dividing ratio. In Table.2, the calculated characteristic impedances of the hybrid from the characteristic impedances of the coupled-line are shown.

Fig. 4 shows the simulation results of the proposed structure with the coupled-line. The switches, 1 ~ 5 in Fig. 2 are turned off at this time. On the other hand, Fig. 5 shows that the simulation results when the switches, 1 ~ 3 and 5 are turned on. In Fig.6 and 7, the simulation results are shown with the same power ratios, 1:1 and 1:0, respectively, for the power divider with hybrids transformed from the coupled-lines.

Table. 1 Characteristic impedances of a coupled-line and a power divider at given power ratios.

	Power divider		Coupled-line	
	1:1	1:0	Z_{oe}	Z_{oo}
Z_1	70.7	50	109	32.45
Z_2	70.7	0	109	32.45

Table. 2 Characteristic impedances of the coupled-line and the hybrid.

Coupled-line		Hybrid	
Z_{oe}	Z_{oo}	Z_1	Z_2
109	32.45	92.362	50.02

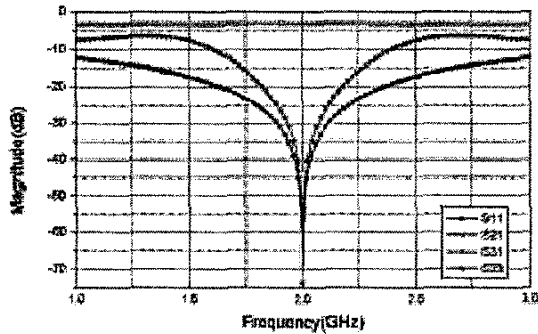


Fig. 4 Simulation results with power ratio of 1:1 (Coupled-line structure)

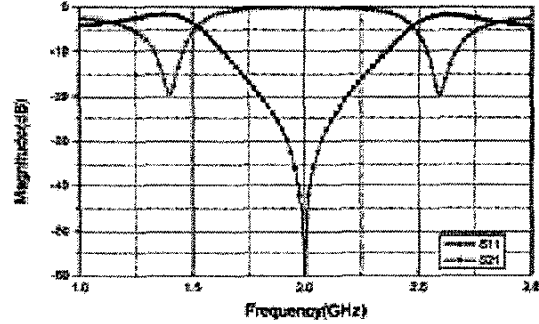


Fig. 5 Simulation results with power ratio of 1:0 (Coupled-line structure)

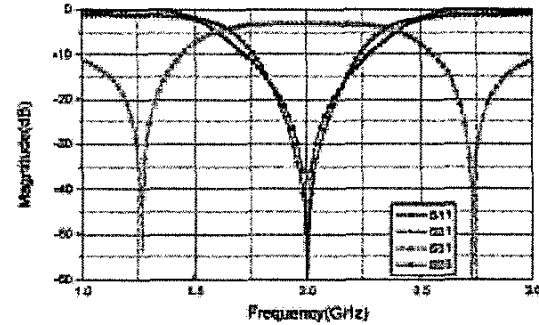


Fig. 6 Simulation results with power ratio of 1:1 (Hybrid structure)

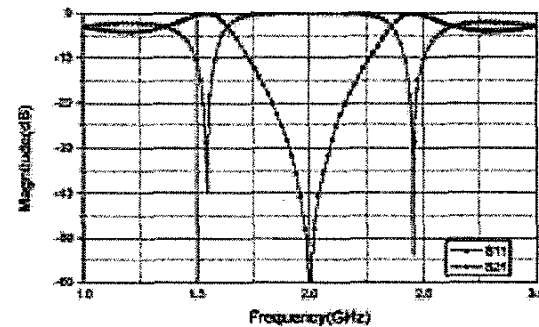


Fig. 7 Simulation results with power ratio of 1:0 (Hybrid structure)

The implemented power divider on the RT/Duroid substrate is shown in Fig. 8. The measured S-parameters of the implemented power divider with the hybrids are shown in Fig. 9 and 10. In Fig. 9, the power divider works as 3 dB power divider. Fig. 10 clearly shows that all input power is delivered to the port 2. The measured data well

agree with simulation results. In Table. 3, the measured results at 2 GHz were summarized.

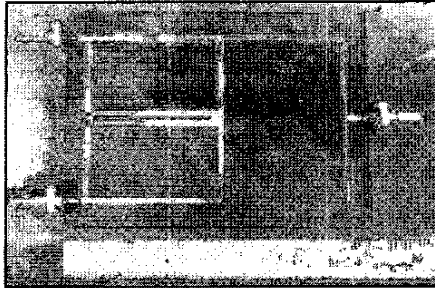


Fig. 8 The implemented power divider

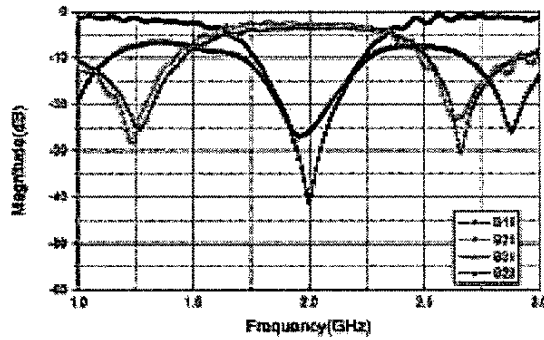


Fig. 9 Measured results with power ratio of 1:1
(Hybrid structure)

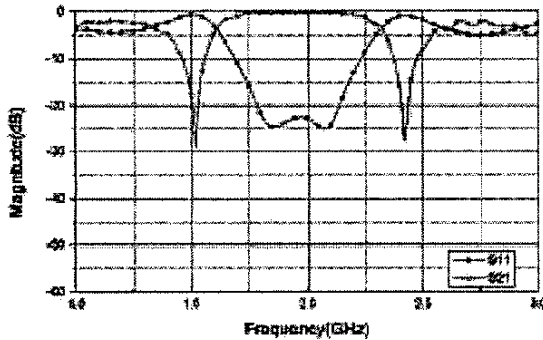


Fig. 10 Measured results with power ratio of 1:0
(Hybrid structures)

Table. 3 Transmitting values at 2GHz with variable power ratios. ($f_0=2\text{GHz}$)

Ratio	S_{21}	S_{31}	S_{23}
1:1	-3.125dB	-3.17dB	-27.84dB
1:0	-0.21dB	none	none

IV. CONCLUSION

In this paper, new power dividing/combining scheme with two different power ratios was proposed. The coupled-line and the hybrid with control-switches were used in the proposed scheme. The measured results well agree with the simulation results. Reconfigurable power divider with 1:1 or 1:0 power dividing ratio was successfully implemented at the frequency of 2 GHz.

ACKNOWLEDGEMENT

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